Data-free Neural Representation Compression with Riemannian Neural Dynamics

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- We view neural nets as dynamical systems: our weights are interactions between neuron states.
- We move beyond Euclidean metrics to capture richer cross-dimensional relations.
- We introduce our RieM to enable parameter-efficient, data-free compression across CNNs, ViTs, and DETR.

Abstract: We model neuron interactions via Riemannian metrics to reconstruct weights without data, achieving superior accuracy–size trade-offs.

Method: RieM (Riemannian Neural Metrics)

- We interpret $W \in R^{m\times n}$ using neuron states $q(in)_x$, $q(out)_y \in R^{Dq}$; we reconstruct weights from neuron interactions.
- We use $D\mu$ sub-metrics $g^{(\alpha)}$ with learnable merges to map interactions to scalars.
- We design two variants: (1) we apply an FFN over concatenated states; (2) we use displacement-based interactions that explicitly mix dimensions.
- We vectorize displacement interactions with a displacement matrix for parallel efficiency.
- We observe that Option Two yields higher nonlinearity and lower fitting error at equal parameters.

Figure placeholder: We compute RieM via displacement vectors, a non-linear merge, and a scalar projection to approximate W_{xy}.

Compression, Inference, and Implementation

- We optimize neuron states and metric parameters by fitting original weights with L2 loss.
- We reduce params from mn to $Dq(m+n) + D\mu(1 + Ds\cdot Dq)$, and we concatenate related layers guided by Shared Correlation Counts.
- We accelerate inference by viewing W as a distance matrix and by dyMerge: we merge similar neurons using learned correlations.
- We quantize neuron states and metric outputs; we tune Dq and merge thresholds to match storage and FLOPs budgets.
- We run 10 random seeds, select the top-3 on dev, and report test averages; we train on a single RTX 4090.

Results: ImageNet and Beyond

Table 1 (ImageNet Quantization): ResNet-18/50 — We show RieM with 8/16-bit achieves higher Top-1 at the same storage than DFQ, UDFC, SQuant, and Lp-Norm baselines; often competitive with fine-tuned methods.

Table 2 (ImageNet Pruning): ResNet-34/101 — We attain higher Top-1 with lower size/FLOPs vs. Neuron Merging and UDFC at 10% and 30% prune ratios.

- We observe strong ViT/Swin performance at 4/8 and 8/8 bits without ViT-specific tailoring.
- We observe on COCO/DETR that we maintain AP with substantial size reductions, outperforming tensor and SVD/quant baselines at equal size.
- We achieve: +0.18% at 8.5× smaller (LeNet-5/MNIST); −0.2% at 4× smaller (Swin-T/ImageNet-1k); +0.1 AP_S at 4.8× smaller (ResNet-DETR/COCO).

Related Work and Open Questions

- We situate our work among data-free quantization (DFQ, ZeroQ, IntraQ, DSG), data-free pruning (parameter pruning, Neuron Merging), and unified data-free compression (UDFC).
- We differ by reconstructing weights from neuron interactions with Riemannian metrics.

Open Questions

- We ask: how can we further refine RieM's analytical form to improve expressivity and stability?
- We ask: can we reduce conversion time while matching target storage/FLOPs automatically?
- We ask: which physics-inspired metrics (e.g., skew/anti-symmetric forms) best trade off accuracy and efficiency?
- We ask: how do dyMerge thresholds interact with accuracy for diverse architectures and datasets?
- We ask: can we extend RieM to sequential or multimodal settings without data?